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Inner Plexiform Layer Substrata Are Discernible with Commercial OCT and Affected by Aging

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Purpose: This study aims to evaluate the inner plexiform layer (IPL) microstructure and its changes with aging using commercial spectral-domain OCT macular scans of healthy individuals with a semiautomated segmentation program.

Design: Cross-sectional study conducted at the Athens Vision Eye Institute from January to July 2024.

Participants: The study included 92 healthy participants.

Methods: OCT images were captured with the Optovue Avanti SD-OCT and processed using ImageJ to measure thickness and analyze the hyperreflective and hyporeflective bands within the IPL. Information about signal intensity, microstructure, and contrast between these sublayers was obtained. Statistical analyses, including Spearman correlation and linear regression, assessed relationships between age and IPL extracted features. Intraeye and intereye repeatability were evaluated using paired samples *t* tests combined with bootstrap analyses.

Main Outcome Measures: The primary outcomes measured were signal intensity of the IPL, contrast between its hyperreflective and hyporeflective bands, and the percentage of IPL with identifiable sublayers. The secondary outcomes included inner retinal thickness measurements, including the IPL, nerve fiber layer (NFL), and ganglion cell complex (GCC).

Results: The IPL exhibited a multilayered structure with 5 sublayers, 3 hyperreflective and 2 hyporeflective, arranged in an alternating pattern. Aging was associated with higher signal intensity from hyporeflective bands and minimal changes in hyperreflective bands, resulting in an overall reduced contrast between the 5 sublayers. Older participants showed a lower percentage of IPL with identifiable sublayers, along with a lower contrast variance within the IPL. Aging also correlated with reduced inner retinal thickness, including the IPL, NFL, and GCC, with a stronger association for the IPL. Inner plexiform layer analysis exhibited high intraeye and intereye repeatability, with significant correlations and nonsignificant mean differences observed in most key parameters.

Conclusions: Analysis of the IPL and its sublayers is both feasible and reproducible using commercially available OCT along with a semiautomated segmentation program. Our findings indicate that the IPL microstructure changes with aging. A comprehensive evaluation of the IPL could serve as a valuable biomarker for early diagnosis and monitoring of diseases affecting synaptic health in this layer.

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Synapses are junctions between neurons that enable signal transmission, playing a key role in sensation, memory, and behavior. To facilitate coordinated communication between neurons, synapses must be precisely organized within neural circuits. The retina provides a remarkable example of this meticulous synaptic arrangement, particularly within the inner plexiform layer (IPL). Located in the inner retina, the IPL is found between the inner nuclear layer, which contains the cell bodies of bipolar cells, and the ganglion cell layer, which contains the ganglion cell bodies. Inner plexiform layer consists entirely of synapses formed by the axons of bipolar cells and dendrites of ganglion and amacrine cells. Through these synapses, IPL integrates and processes visual signals coming from the photoreceptors through the bipolar cells before transmitting them to the brain via the retinal ganglion cells (RGCs).[1]

Within the IPL, synapses are further organized histologically into 5 strata: the 2 innermost strata correspond to

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sublamina B; the 2 outermost to sublamina A; and the middle stratum acts as a watershed zone. This stratification, first noted in histology by Santiago Ramón y Cajal in 1893,[2] has been confirmed by various studies[3–7] and has both structural and functional significance, with different neuronal subtypes terminating in specific sublaminae. However, most knowledge of this organization comes from animal studies, with limited human data primarily from postmortem histology.[8]

Aging negatively impacts synaptic health, and synaptic dysfunction contributes to various eye diseases, like glaucoma, and neurodegenerative disorders, including Alzheimer disease and Parkinson disease.[9–12] Advanced imaging technologies, particularly spectral-domain OCT, offer high-resolution, cross-sectional imaging of the human retina, providing a unique opportunity for in vivo noninvasive assessment of human synapses. The retina's laminar organization, characterized by alternating synaptic and nuclear layers, creates varying levels of reflectivity that form OCT imaging pattern.[13,14] However, the subtle reflectivity changes that differentiate the IPL sublayers are less pronounced than those between the nuclear and synaptic layers. Only a few studies using mostly experimental machines have successfully identified IPL sublayers showing 3 hyperreflective bands alternating with 2 hyporeflective bands.[15–19] Distinguishing these sublayers often requires depth resolution beyond that of most commercial OCT systems, requiring the use of advanced technologies.[15–18]

In this study, we employed high-resolution scans from commercial-band OCT alongside semiautomated image analysis program to analyze the IPL and its 5 sublayers. We report IPL thickness and reflectivity patterns in a cohort of healthy human subjects, investigating how aging affects IPL thickness and stratification on OCT.

Methods

A flow diagram of the study, detailing all the steps involved, from participant selection and ophthalmic examination to OCT imaging, IPL analysis, and the assessment of intraeye and intereye repeatability is shown in Figure 1.

Participant Selection

This cross-sectional study involved healthy individuals examined at the Athens Vision Eye Institute in Greece from January to July 2024. We excluded participants with retinal pathology, unexplained vision loss, prior eye trauma or retinal surgery, and medications affecting the retina (e.g., chloroquine). Those with glaucoma or systemic conditions that could impact vision, such as diabetes and uncontrolled hypertension, were also excluded. Additional exclusion criteria included a refractive spherical equivalent of ±6.00 diopters (D) or greater, astigmatism of ±3.00 D or greater, and intraocular pressure exceeding 21 mmHg. Participants with significant cataracts (nuclear sclerosis grade +2 or higher) or other media opacities affecting imaging quality were also excluded.

Both eyes were assessed, and the eye with the best-quality OCT image, based on signal-to-noise ratio, was included in the main cohort. The other eye was included in the intereye repeatability analysis to assess consistency between the eyes of an individual, as well as in the intraeye repeatability analysis to assess consistency between measurements of the same eye.

The study adhered to the Declaration of Helsinki and received approval from the Athens Vision Institutional Review Board, with written informed consent obtained from all eligible participants.

Ophthalmology Examination

Participants provided demographic information and medical histories. A trained ophthalmologist, blinded to the study's objectives, conducted comprehensive eye examinations, including best-corrected visual acuity assessment using a Snellen chart. Intraocular pressure was measured using Goldmann applanation tonometer. Anterior biomicroscopy and dilated fundus examination using 0.5% tropicamide were performed. Optical biometry with measurement of axial length using the IOLMaster 500 (Carl Zeiss Meditec AG) was available in a small random subset of participants.

OCT Imaging

An experienced operator, blinded to participants' characteristics, performed OCT scans on the same day as the other measurements using the Optovue Avanti SD-OCT (Visionix or Optovue Inc). Participants underwent several scan protocols, including macula raster, enhanced high definition (HD) scan, retina map, and a glaucoma protocol for retinal nerve fiber layer (RNFL) and RGC analysis. The enhanced HD scan was utilized for IPL analysis due to its superior resolution capabilities. This 12-mm widefield scan provides high-resolution imaging of the retina with an axial resolution of 5 μ and a transverse resolution of 15 μ. The macula raster scan confirmed the absence of retinal pathology, while the retina map derived inner and outer retina thickness measurements. Retinal nerve fiber layer and RGC analyses provided measurements for NFL and ganglion cell complex thicknesses. Poor quality images—evidenced by a signal-to-noise ratio below 40, movement artifacts causing misalignment, breaks, and localized weak signals due to artifacts such as floaters, or an IPL not clearly visible in the area of interest—were excluded before IPL analysis.

IPL Analysis

Inner plexiform layer analysis was conducted as illustrated in Figure 2. Enhanced HD scans were imported into ImageJ.[20] To enhance image quality for IPL data extraction, we applied linear interpolation, which is a method that enlarges images by estimating intermediate pixel values and adjusted local contrast using the contrast limited adaptive histogram equalization algorithm. This algorithm enhances image contrast by dividing the image into small tiles, applying histogram equalization locally, and clipping histograms to limit noise amplification. Tiles are reassembled using bilinear interpolation, ultimately improving local contrast while preserving global coherence. The IPL region of interest was selected manually and spanned from 1000 μ distal to the foveal center, where the IPL visually exhibited 5 distinct sublayers, to 2500 μ distal to foveal center, approaching peripapillary region where the sublayers became less discernible. Although 5 sublayers could occasionally be

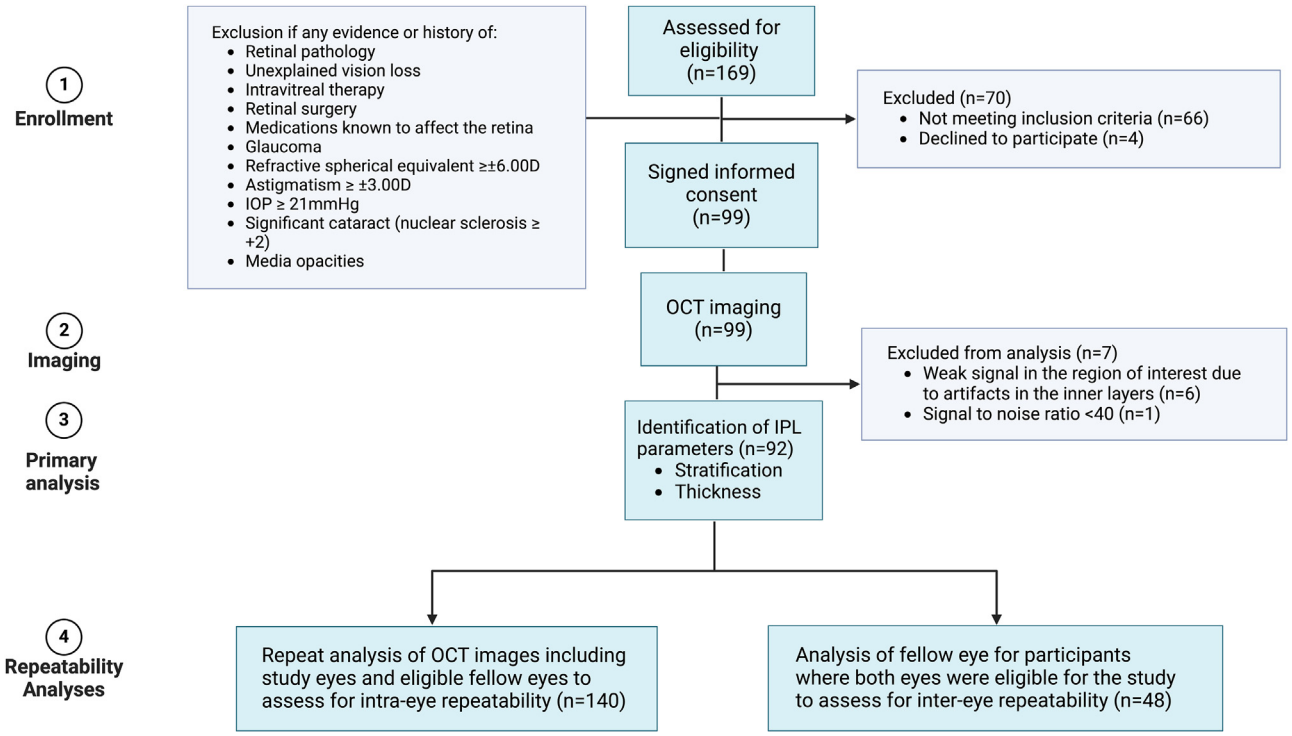


Figure 1. Flow diagram of the study. The figure illustrates the methodology of the cross-sectional study. A total of 169 individuals (both eyes) were screened for eligibility. Out of these, 99 individuals were deemed eligible (with at least 1 eligible eye), provided informed consent, and underwent OCT imaging. Analysis of the IPL and its sublayers was successfully conducted in 92 of these individuals. For participants with both eligible eyes, the eye with the best OCT image quality (determined by the highest signal-to-noise ratio) was included in the main analysis, while the fellow eye was used for intereye repeatability assessment to evaluate consistency between eyes. The eligible eyes (92 from the main analysis and the corresponding fellow eyes) were subsequently analyzed a second time to assess intraeye repeatability, ensuring the consistency of results from repeat analyses of the same eye. D = diopters; IOP = intraocular pressure; IPL = inner plexiform layer; n = number. Created in BioRender. Emfietzoglou, M. (2025) https://BioRender.com/c39x641.

identified beyond these limits in some eyes, a standardized region of interest (1000−2500 μ from the foveal center toward the optic disc) was consistently applied across all images to ensure methodological uniformity.

Inner plexiform layer thickness was measured manually across the region of interest to obtain an average value. Pixel profile lines were generated using 15-μ horizontal intervals across the IPL segment. Extracted pixel values were normalized using a z score transformation to standardize intensity values across images, allowing for direct comparison, and analyzed in Python to identify hyperreflective and hyporeflective bands, as well as to extract features such as sublayer mean pixel values and relative distances between the bands (accounting for differences in IPL thickness) and adjacent sublayer contrast.

To address contrast artifacts arising from minor IPL depth variations and sublayer undulation within the papillomacular bundle, we analyzed the IPL in 300-μ segments. This approach allowed us to create a contrast-adjusted IPL pixel profile that more accurately represents the true longitudinal mean profile. The percentage of these segments for which the 5 sublayers could be identified was expressed as the percentage of IPL segments with identifiable sublayers.

A linear regression was fitted to the IPL sublayers pixel profile and the mean squared error (MSE) was calculated. A higher MSE indicates greater contrast between adjacent sublayers, suggesting improved IPL segmentation. All image segmentations were performed by an individual blinded to participant characteristics.

Statistical Analysis

We performed statistical analysis using SPSS version 29.0.2.0 (SPSS Inc). Descriptive statistics were calculated to summarize the demographic, ocular, and OCT characteristics of the participants. Continuous variables were expressed as medians with interquartile ranges or means with standard deviations, as appropriate. Categorical variables were reported as frequencies and percentages. We used Spearman rank correlation coefficient to assess the relationship between age and various OCT parameters, given the nonparametric nature of the data. Linear regression analyses were conducted to further explore these associations while adjusting for confounding factors. To account for multiple comparisons, P values were adjusted using the Bonferroni correction, establishing a threshold of 0.0024 (P value of 0.05 divided by 21 comparisons) for statistical significance.

For all available eyes, including fellow eyes of participants where both eyes were eligible, we assessed intraeye repeatability (consistency of measurements obtained from the same eye) by repeating the IPL analysis and conducting a paired samples t tests in conjunction with bootstrap analysis. For each IPL parameter analyzed, we calculated correlation coefficients to determine the strength of the relationship between the repeated measurements and computed mean differences and standard errors to quantify the average deviations. A Bonferroni correction was applied to account for multiple comparisons. We also reported Cohen d and Hedges g (corrects for small sample sizes) as measures of effect size. Effect sizes were categorized as small (0.2), medium (0.5), and large

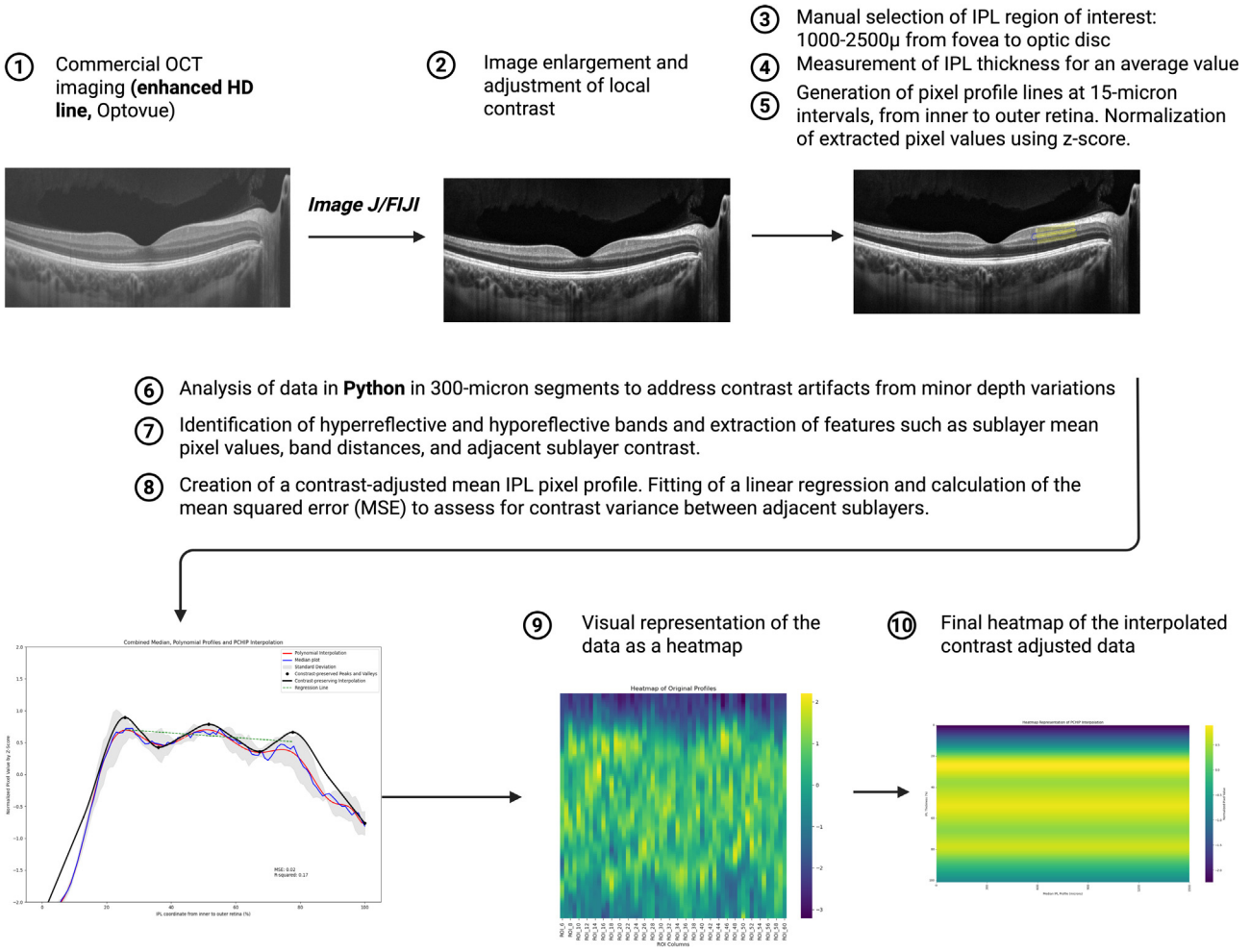


Figure 2. Steps involved in IPL analysis. This figure outlines the methodology for processing OCT images of the IPL. OCT images were imported into ImageJ, adjusted for aspect ratio and resolution, and the region of interest was manually selected and defined between 1000 and 2500 µ from the foveal center to the optic disc. Inner plexiform layer thickness was measured multiple times across this area and the average value was calculated. Pixel profile lines were generated at 15-µ intervals across the IPL segment, from the inner to outer retina. To address contrast artifacts due to minor IPL depth variations within the papillomacular bundle, the IPL was analyzed in 300 µ segments, creating a contrast-adjusted pixel profile that accurately reflects the true longitudinal mean profile. A linear regression was fitted to the IPL sublayers pixel profile, with mean squared error calculated to assess contrast between adjacent sublayers; a higher mean squared error indicates improved IPL segmentation. The illustrated example has an MSE of 0.02; values in the sample ranged from 0 to 0.12 (median: 0.01, IQR: 0.01). All segmentations were performed by an individual blinded to participant characteristics. IPL = inner plexiform layer; PCHIP = piecewise cubic hermite interpolating polynomial. Created in BioRender. Emfietzoglou, M. (2025) https://BioRender.com/b26z478.

(≥0.8), reflecting minor, noticeable, and substantial differences, respectively. This comprehensive approach allowed for a thorough evaluation of intraeye repeatability in our study. A similar analysis was conducted to assess intereye repeatability (consistency of measurements between the eyes of the same participant), including both eyes of individuals where both eyes were eligible for the study, further validating the reliability of the IPL analysis.

Results

Descriptive Statistics

We included 92 participants with a median age of 57 and a balanced gender ratio. All participants exhibited clear optical media, resulting in high-quality OCT images, with a median signal-to-noise ratio of 59. Table 1 presents the descriptive characteristics of the study subjects. Table 2 outlines the descriptive statistics of the parameters obtained from the OCT analysis. A typical stratification pattern consisting of 2 hyporeflective and 3 hyperreflective bands was identified in all participants. Figure 3 illustrates the average intensity profile of our study population, showcasing a distinct pattern with 3 hyperreflective and 2 hyporeflective bands, and provides explanations for the derived parameters.

Aging and IPL

Table 3 summarizes the results of Spearman rho (ρ) and linear regression analyses examining the relationships

Table 1. Descriptive Characteristics of Study Participants (n = 92)

Descriptive characteristics	Summary Statistics
Demographic and Systemic Characteristics	
Caucasian	92 (100%)
Age, yrs, median	57 (19.5)
Gender, female	48 (52%)
Hypertension, n	18 (20%)
Dyslipidemia, n	26 (28%)
Ocular characteristics	
Study eye, OD	46 (50%)
Lens status, n	
Phakic	84 (91%)
Clear	69
Early nuclear sclerosis	15
Pseudophakic	8 (9%)
Axial length (mm)*	23.8 (1.49)
Spherical equivalent, D	0.00 (2.25)
BCVA, LogMAR	0.00 (0)
IOP (mmHg)	14.7 (2.2)
SNR value of OCT image	59 (15)

BCVA = best-corrected visual acuity; D = diopter; IOP = intraocular pressure; IQR = interquartile range; LogMAR = logarithm of the minimum angle of resolution; n = number; OD = right eye; SD = standard deviation; SNR = signal-to-noise ratio.
The data are expressed as median (IQR), mean (SD), or N (%), as appropriate.
*Axial length was available only in a subset of patients (31 participants, 46 eyes).

Table 2. Descriptive Statistics of OCT Parameters (n = 92)

OCT Parameter	Median (IQR)
IPL Analysis	
Signal intensity	
Inner hyperreflective	0.92 (0.28)
Middle hyperreflective	0.88 (0.19)
Outer hyperreflective	0.77 (0.16)
Inner hyporeflective	0.62 (0.35)
Outer hyporeflective	0.49 (0.23)
Relative distances between adjacent bands	
Inner hyperreflective, inner hyporeflective	10.90 (3.62)
Inner hyporeflective, middle hyperreflective	12.50 (5.62)
Middle hyperreflective, outer hyporeflective	13.71 (4.00)
Outer hyporeflective, outer hyperreflective	9.71 (3.65)
Contrast between adjacent bands	
Inner hyperreflective, inner hyporeflective	0.30 (0.22)
Inner hyporeflective, middle hyperreflective	0.28 (0.24)
Middle hyperreflective, outer hyporeflective	0.36 (0.19)
Outer hyporeflective, outer hyperreflective	0.25 (0.20)
% of IPL segments with identifiable sublayers	55 (25)
MSE	0.01 (0.01)
Retinal layer(s) thickness (μ)	
Inner retina	133 (15.5)
Outer retina	190 (19)
IPL	39 (4)
NFL	97 (13.75)
GCC	98 (10)

GCC = ganglion cell layer complex; IPL = inner plexiform layer; MSE = mean squared error; NFL = nerve fiber layer.
The data are expressed as median (IQR). Inner retina contains the nerve fiber layer, ganglion cell layer, IPL, and inner nuclear layer; outer retina contains the outer nuclear layer, outer plexiform layer, and the photoreceptors. Inner plexiform layer was measured manually as part of the OCT analysis on ImageJ 1000 μ to 2500 μ from the fovea to the optic disc. Inner retina and outer retina thicknesses refer to the nasal parafoveal region; NFL and GCC thicknesses are average measurements from all macular sectors obtained from the built-in Optovue automated segmentation program. Signal intensity and contrast between bands are described as z score normalized pixel values.

between age and IPL parameters. To determine which variables to include as confounders, we assessed potential confounders and found significant relationships (P values adjusted for Bonferroni correction, threshold of 0.0024) between image quality and several IPL metrics, including the percentage of IPL with 3 hyperreflective and 2 hyporeflective bands, the signal from the hyporeflective bands, the contrast between the hyperreflective and the hyporeflective bands, and the MSE. Due to the importance of image quality for reliable analysis, we opted to adjust for image quality in all our regression models.

Our results show that age correlates with increased signal from the hyporeflective bands, with outer hyporeflective band remaining significant even after Bonferroni correction and adjustment for image quality. Conversely, age does not affect the signal from the hyperreflective bands, resulting in an overall decrease in contrast between the bands. The association for the contrast between the middle hyperreflective band and the inner hyporeflective band does not retain significance after the Bonferroni correction and the adjustment for image quality. Additionally, we found no association between age and the relative distance between the IPL sublayers.

As individuals age, the percentage of IPL segments showing 5 distinct bands decrease. Older age also correlates with lower MSE values, indicating a flatter curve and less distinct segmentation within the IPL. In contrast, younger individuals exhibit higher MSE values, reflecting clearer IPL sublayers. This association remains significant after adjusting for image quality. Figure 4 illustrates the age-related changes in IPL intensity profiles between individuals younger and older than 60 years. Additionally, aging is associated with a reduced inner retinal thickness, including thinner IPL, NFL, and ganglion cell complex, with a stronger association observed between aging and IPL compared to the other layers.

Intraeye Repeatability

We repeated the IPL analysis in 140 eyes of 92 participants. Table 4 presents the results of the paired samples t test with bootstrap analysis to evaluate intraeye repeatability. Most parameters, especially the contrast between adjacent hyperreflective and hyporeflective bands, showed strong correlations, indicating high repeatability. Most parameters had minimal mean differences that were not significant, except for the contrast difference between the middle hyperreflective band and the inner hyporeflective band, which had a significant mean difference but a small effect size. We observed substantial effect sizes for the relative distances between the IPL bands, exceeding 3, suggesting notable differences in repeated measurements; however,

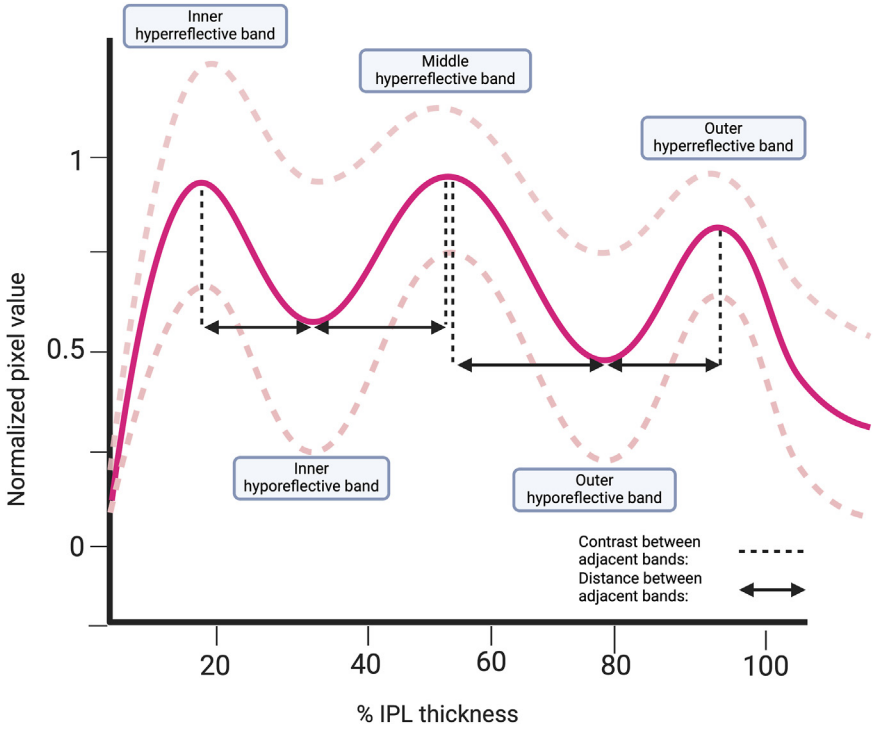


Figure 3. The median IPL intensity profile. This figure shows a representative example of the IPL intensity profile plotted against the percentage of IPL thickness. The profile displays a distinct pattern with 3 hyperreflective and 2 hyporeflective bands, indicating variations in signal intensity throughout the IPL. The signal intensity of each hyperreflective and hyporeflective band approximates the median values of the study population (pink profile curve). The dotted light-pink profiles represent the IQR values. Dotted lines illustrate the contrast between the adjacent hyperreflective and hyporeflective bands, while arrows denote the relative distances between them. IPL = inner plexiform layer, IQR = interquartile range. Created in BioRender. Emfietzoglou, M. (2025) https://BioRender.com/k48a568.

these differences were not significant, likely due to high variability in their measurements.

Intereye Repeatability

We analyzed the IPL in the fellow eye of 48 participants with both eyes eligible. Table 5 presents the results of the paired samples *t* test with bootstrap analysis to evaluate intereye repeatability. Significant correlations were found in key parameters, especially in the contrast between the hyperreflective and hyporeflective bands, the MSE, and the percentage of IPL with 3 hyperreflective and 2 hyporeflective bands, indicating high intereye consistency. These parameters showed nonsignificant mean differences and small effect sizes. The remaining measurements had moderate to minimal correlations, with no significant mean differences, and most displayed small effect sizes, confirming their reliability. However, the relative distances between IPL bands had substantial effect sizes, indicating measurement variability.

Discussion

To our knowledge, no studies have investigated the impact of aging on IPL stratification in humans. We demonstrated in a group of healthy individuals that the IPL has a typical intensity profile with 3 hyperreflective and 2 hyporeflective bands; aging is linked to higher signal from hyporeflective bands, while hyperreflective bands stays relatively stable, leading to an overall decrease in contrast between the bands; there is no significant relationship between age and relative distances between the bands; the percentage of the IPL segments where the 3 hyperreflective and 2 hyporeflective bands can be identified decreases with aging; and aging is associated with reduced inner retina thickness, including a thinner IPL.

The IPL intensity profile in our study, featuring 3 hyperreflective and 2 hyporeflective bands, corroborates with previous research. Most prior studies used advanced, noncommercial OCT technologies—such as broadband light sources and adaptive optics—to achieve the necessary image quality for visualizing IPL sublayers.[15–18] Only 1 prior study used a commercial OCT system with A-scan rates of 20 kHz and 85 kHz via Heidelberg Engineering's SHIFT technology and identified the IPL sublayers in 40% of patients at 85 kHz and in 78.3% of patients at 20 kHz. However, the slower rate's longer acquisition time limits its clinical applicability, particularly for eyes with pathology, decreased visual acuity, or unstable fixation, and may introduce artifacts.[19] In contrast, we used standard commercial OCT technology from Optovue at a standard rate of 70 kHz along with a semiautomated image analysis

Table 3. Spearman Rho and Linear Regression Models for Age and IPL Features (n = 92)

Signal Intensity				
Inner Hyperreflective	*Middle Hyperreflective*	*Outer Hyperreflective*	*Inner Hyporeflective*	*Outer Hyporeflective*
Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)
−0.14 (0.17)	0.06 (0.57)	0.19 (0.06)	0.21 (0.04)	**0.43 (<0.001)**
Univariate (*P*)	Univariate (*P*)	Univariate (*P*)	Univariate (*P*)	Univariate (*P*)
−1.27 (0.23)	0.79 (0.46)	0.14 (0.18)	0.20 (0.05)	**0.40 (<0.001)**
Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)
−1.83 (0.11)	0.07 (0.12)	0.15 (0.21)	0.10 (0.37)	**0.36 (<0.001)**
Relative distances between adjacent bands				
Inner hyperreflective, inner hyporeflective	*Inner hyporeflective, middle hyperreflective*	*Middle hyperreflective, outer hyporeflective*		*Outer hyporeflective, outer hyperreflective*
Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)		Spearman ρ (*P*)
−0.21 (0.84)	0.10 (0.33)	−0.01 (0.95)		−0.24 (0.02)
Univariate (*P*)	Univariate (*P*)	Univariate (*P*)		Univariate (*P*)
0.07 (0.49)	0.06 (0.56)	−0.01 (0.94)		−0.17 (0.12)
Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)		Adjusted (*P*)
0.10 (0.41)	0.07 (0.53)	−0.03 (0.82)		−0.12 (0.28)
Contrast between adjacent bands				
Inner hyperreflective, inner hyporeflective	*Inner hyporeflective, middle hyperreflective*	*Middle hyperreflective, outer hyporeflective*		*Outer hyporeflective, outer hyperreflective*
Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)		Spearman ρ (*P*)
−0.39 (<0.001)	−0.26 (0.01)	**−0.43 (<0.001)**		**−0.43 (<0.001)**
Univariate (*P*)	Univariate (*P*)	Univariate (*P*)		Univariate (*P*)
−0.38 (<0.001)	−0.22 (0.03)	**−0.39 (<0.001)**		**−0.43 (<0.001)**
Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)		Adjusted (*P*)
−0.32 (0.003)	−0.13 (0.23)	**−0.40 (<0.001)**		**−0.37 (<0.001)**
Contrast variance within the IPL				
% of IPL segments with identifiable sublayers				*MSE*
Spearman ρ (*P*)				Spearman ρ (*P*)
−0.48 (<0.001)				**−0.44 (<0.001)**
Univariate (*P*)				Univariate (*P*)
−0.44 (<0.001)				**−0.41 (<0.001)**
Adjusted (*P*)				Adjusted (*P*)
−0.34 (<0.001)				**−0.36 (<0.001)**
Retinal layer(s) thickness				
Inner retina	*Outer retina*	*IPL*	*NFL*	GCC
Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)	Spearman ρ (*P*)
−0.29 (0.001)	0.12 (0.27)	**−0.38 (<0.001)**	**−0.46 (<0.001)**	**−0.31 (0.004)**
Univariate (*P*)	Univariate (*P*)	Univariate (*P*)	Univariate (*P*)	Univariate (*P*)
−0.25 (0.02)	0.17 (0.12)	**−0.46 (<0.001)**	**−0.40 (<0.001)**	−0.26 (0.02)
Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)	Adjusted (*P*)
−0.22 (0.07)	0.12 (0.31)	**−0.59 (<0.001)**	**−0.43 (<0.001)**	−0.30 (0.01)

GCC = ganglion cell layer complex; IPL = inner plexiform layer; MSE = mean squared error; NFL = nerve fiber layer.
The table presents the correlation and regression analysis results for age and various IPL features, including contrast and relative distance between adjacent sublayers, IPL contrast variance, and retinal layer thicknesses. Spearman rank correlation coefficients indicate the strength and direction of the correlation, while *P* values assess statistical significance. Univariate and adjusted regression results (adjusted for image quality) are also provided for each parameter, with corresponding *P* values. The level of significance was corrected using Bonferroni for multiple comparisons (21 tests), resulting in a threshold *P* value of 0.05 of 21 ≈ 0.0024. Features with statistically significant associations with aging are highlighted in bold.

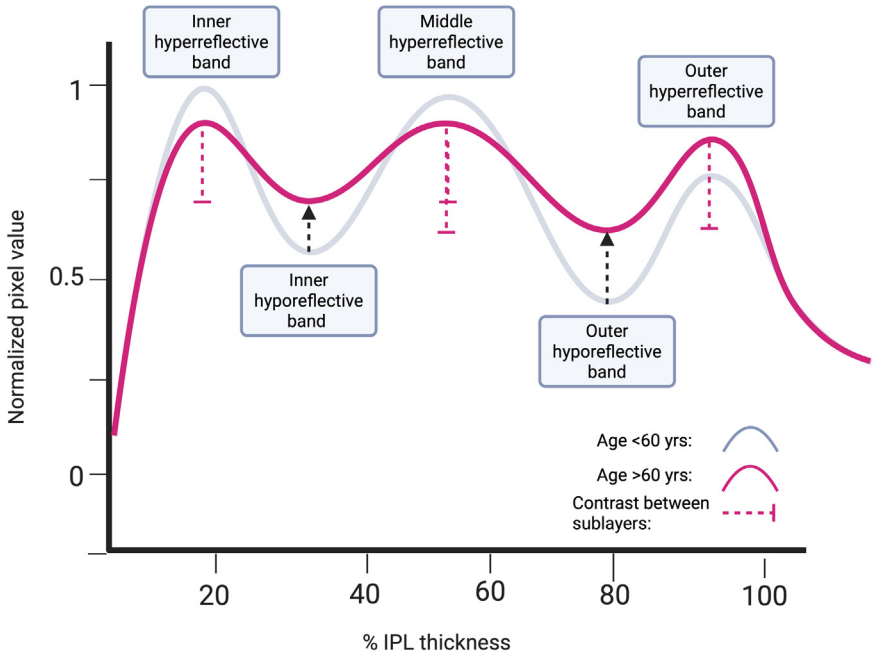


Figure 4. Age-related changes in IPL intensity profile. This schematic representation illustrates how aging affects the intensity profile of the IPL. The light blue curve represents the median intensity profile of individuals <60 years of age, while the pink curve depicts the median profile of individuals >60 years of age. Dotted arrows show the changes observed with increasing age. Aging is associated with prominent alterations in the signal intensity of hyporeflective bands and minimal changes in the signal of the hyperreflective bands, leading to an overall decrease in contrast between adjacent bands and, ultimately, a flatter intensity curve. IPL = inner plexiform layer. Created in BioRender. Emfietzoglou, M. (2025) https://BioRender.com/s09c751.

Table 4. Assessment of Intraeye Repeatability (n = 140)

OCT Parameter	Correlation (*P*)	Mean Difference (*P*)	Cohen d	Hedge Correction
Signal intensity				
Inner hyperreflective	0.90 (<0.001)	−0.02 (0.08)	0.11	0.11
Middle hyperreflective	0.55 (<0.001)	−0.01 (0.56)	0.12	0.12
Outer hyperreflective	0.50 (<0.001)	−0.02 (0.05)	0.13	0.13
Inner hyporeflective	0.83 (<0.001)	−0.03 (0.02)	0.14	0.14
Outer hyporeflective	0.68 (<0.001)	−0.02 (0.12)	0.16	0.16
Relative distances between adjacent bands				
Inner hyperreflective, inner hyporeflective	0.23 (0.003)	0.19 (0.55)	**3.77**	**3.79**
Inner hyporeflective, middle hyperreflective	0.49 (<0.001)	0.45 (0.15)	**3.73**	**3.75**
Middle hyperreflective, outer hyporeflective	0.36 (<0.001)	0.23 (0.49)	**3.93**	**3.95**
Outer hyporeflective, outer hyperreflective	0.30 (<0.001)	−0.56 (0.09)	**3.85**	**3.87**
Contrast between adjacent bands				
Inner hyperreflective, inner hyporeflective	0.88 (<0.001)	0.02 (0.06)	0.10	0.10
Inner hyporeflective, middle hyperreflective	0.85 (<0.001)	0.03 (0.001)	0.09	0.09
Middle hyperreflective, outer hyporeflective	0.80 (<0.001)	0.02 (0.08)	0.11	0.10
Outer hyporeflective, outer hyperreflective	0.80 (<0.001)	0.00 (0.67)	0.10	0.10
% of IPL segments with identifiable sublayers	0.65 (<0.001)	0.02% (0.10)	0.16%	0.16%
MSE	0.85 (<0.001)	0.00 (0.51)	0.01	0.01

IPL = inner plexiform layer; MSE = mean squared error.
The table presents the results of a paired samples *t* test with bootstrap analysis, evaluating the intraeye repeatability of various IPL features. For each feature, correlation coefficients and *P* values indicate the strength and significance of the repeatability. A Bonferroni correction was applied to the *P* values (0.05 of 21) to assess statistical significance for both the correlation between the 2 repeated measurements and the mean differences between them. The mean differences and *P* values provide insights into average deviations between measurements. The same corrected *P* value was used to determine whether these mean differences were statistically significant. Cohen d and Hedges g are reported as measures of effect size to quantify the strength of the differences between groups. Effect sizes are categorized as small (0.2), medium (0.5), and large (0.8 or above), reflecting minor, noticeable, and substantial differences, respectively. Features with large effect sizes and inconsistencies across repeated measurements are highlighted in bold.

Table 5. Assessment of Intereye Repeatability (n = 48, eyes = 96)

OCT Parameter	Correlation (P)	Mean Difference (P)	Cohen d	Hedge Correction
Signal intensity				
Inner hyperreflective	0.26 (0.04)	−0.02 (0.67)	0.25	0.25
Middle hyperreflective	**−0.08 (0.30)**	−0.01 (0.91)	0.12	0.12
Outer hyperreflective	0.37 (0.004)	−0.05 (0.04)	0.14	0.14
Inner hyporeflective	**0.16 (0.13)**	−0.02 (0.69)	0.23	0.23
Outer hyporeflective	0.43 (0.001)	−0.04 (0.22)	0.20	0.21
Relative distances between adjacent bands				
Inner hyperreflective, inner hyporeflective	**0.11 (0.24)**	0.10 (0.87)	**2.88**	**2.93**
Inner hyporeflective, middle hyperreflective	**0.20 (0.09)**	0.50 (0.43)	**3.69**	**3.75**
Middle hyperreflective, outer hyporeflective	**−0.01 (0.47)**	1.13 (0.11)	**3.28**	**3.33**
Outer hyporeflective, outer hyperreflective	**−0.01 (0.47)**	1.31 (0.06)	**3.31**	**3.37**
Contrast between adjacent bands				
Inner hyperreflective, inner hyporeflective	0.62 (<0.001)	0.00 (0.98)	0.21	0.22
Inner hyporeflective, middle hyperreflective	0.38 (0.004)	0.02 (0.32)	0.15	0.15
Middle hyperreflective, outer hyporeflective	0.37 (0.005)	0.04 (0.14)	0.19	0.19
Outer hyporeflective, outer hyperreflective	0.65 (<0.001)	0.00 (0.90)	0.17	0.17
% of IPL segments with identifiable sublayers	0.42 (0.002)	0.04% (0.20)	0.19%	0.19%
MSE	0.73 (<0.001)	0.00 (0.30)	0.02	0.02

IPL = inner plexiform layer; MSE = mean squared error.
The table presents the results of a paired samples *t* test with bootstrap analysis, evaluating the intereye repeatability of various parameters related to IPL measurements. For each parameter, correlation coefficients and *P* values indicate the strength and significance of the repeatability. A Bonferroni correction was applied to the *P* values (0.05 of 21) to assess statistical significance for both the correlation between the 2 repeated measurements and the mean differences between them. The mean differences and standard errors provide insights into average deviations between measurements. The same corrected *P* value was used to determine whether these mean differences were statistically significant. Cohen d and Hedge g are reported as measures of effect size to quantify the strength of the differences between groups. Effect sizes are categorized as small (0.2), medium (0.5), and large (0.8 or above), reflecting minor, noticeable, and substantial differences, respectively. Features with large effect sizes and inconsistent measurements between eyes are highlighted in bold.

program, specifically the enhanced HD line imaging option, which effectively evaluated IPL lamination without needing external light sources.

This reflectivity pattern indicates differences in reflectance within the IPL but does not fully capture the layer's complexity. However, it could align with histological and immunocytochemical studies that traditionally divide the IPL into 5 strata (S1−S5).[21] This stratification is crucial for visual function as specific synapses seem to occur at distinct strata. Ganglion cell dendrites must reside in specific strata to connect with particular amacrine and bipolar neurons.[1] Typically, ON cells, which respond to increases in light, are found in sublamina B (S3−S5), while OFF cells, which respond to decreases, are in sublamina A (S1−S2).[22,23] However, recent connectomic studies of the mouse retina have revealed deviations from this strict stratification, with some ON cells targeting OFF IPL layers, further complicating the interpretation of OCT findings.[24−26]

In our study population, we found that not only IPL thickness decreases with age, but also differences in intensity between the sublayers become less distinguishable. It would be reasonable to consider that this reduced variation may stem from the decreased IPL thickness, which brings the sublayers closer together. However, our intensity profiles are plotted against the IPL's percentage thickness. Additionally, we did not observe any significant changes in the relative distances between adjacent bands.

Our findings align with existing research on the aging retina in animal models, indicating that the overall organization and lamination of the neural retina remain intact over time.[27] Vision decline in the elderly is primarily attributed to dysfunctional retinal neurons, dysregulated energy metabolism, and impaired proteostasis, leading to disrupted communications among these neurons.[28−30] This also applies to the lamellar structure of the IPL, which remains unchanged with aging.[31] The dendrites within the IPL preserve their qualitative features, including molecular identity and laminar specificity. However, there are notable quantitative changes, such as reduced dendritic and axonal arbor areas and decreased synapse density.[31]

Although the reflectivity patterns observed in OCT scans cannot be directly compared to cellular and molecular structures, they are influenced by the underlying characteristics of the retina. The overall hyperreflectivity of the IPL, compared to the adjacent hyporeflective inner nuclear layer, is thought to arise from synaptic structures rich in membranes, vesicles, and mitochondria.[32] Similarly, variations in reflectivity within the IPL may be attributed to the distinct dendritic features of bipolar, amacrine, and ganglion cells, as revealed by electron microscopy. Bipolar cells possess axonal endings filled with synaptic vesicles and exhibit a ribbon-like density, whereas ganglion cell endings lack synaptic vesicles and contain neurotubules, ribosomes, and glycogen granules. Amacrine cell endings also contain synaptic vesicles but are characterized by clustered membrane

densities. Thus, we can hypothesize that the changes observed in OCT scans of older individuals may be related to alterations in these neuronal populations.[33,34]

Our hypothesis is that age-related changes within the IPL are predominantly driven by alterations in amacrine cells. Amacrine cells are a diverse group of inhibitory interneurons located in the inner nuclear layer, with synapses extending into the IPL.[35] They play a crucial role in visual processing by integrating and modulating signals from bipolar cells before transmitting them to ganglion cells. These cells are essential for functions such as dim-light vision, dark adaptation, contrast sensitivity, motion processing, and color perception—all of which decline in the elderly.[36] Previous studies have documented age-related changes in amacrine cells, including loss of AII amacrine cells, which is one of the predominant subtypes. Notably, the dendrites of AII amacrine cells, stratify in S2 and S4 to 5.[28,37] These strata seem to correlate with the hyporeflective bands in which we observed the most pronounced findings on OCT. Additionally, the number of dopaminergic amacrine cells, which primarily stratify in stratum 1, has been shown to reduce by half in the aging retina.[38]

Another type of amacrine cell that may contribute to the observed changes is the starburst amacrine cell, an interneuron that expresses gamma-aminobutyric acid and choline acetyltransferase, with its dendrites stratified in S2 and S4 of the IPL. These cells play a crucial role in motion perception by generating direction-selective responses to moving stimuli. In Parkinson disease, where motion perception disturbances are common, there is a reduction in the density of choline acetyltransferase–positive amacrine cells in the retina, along with a loss of synaptic connectivity between these cells and dopaminergic neurons.[21] In addition, all inner retinal layers, including the IPL, are thinner in Parkinson disease compared to healthy controls.[39]

Recent data indicate that axonal injury—whether from glaucoma or neurodegenerative processes—induces rapid changes in synaptic structures before leading to apoptosis or noticeable loss of RGCs.[40–42] Additionally, different RGC subtypes respond variably to optic nerve injury with specific subtypes, such as those with dendrites in the OFF sublamina, being particularly sensitive.[43] Therefore, analyzing IPL different strata could yield valuable biomarkers for early diagnosis and progression of diseases that could alter these synapses or the vascular supply to these layers, such as glaucoma[44–47] or diabetes mellitus.[48–50] Inner plexiform layer analysis also holds promise for assessing synaptic health and function in neurodegenerative diseases such as Alzheimer and Parkinson—as mentioned above, as OCT provides significantly higher resolution imaging of nervous structures compared to brain magnetic ressonance imaging.[51–53]

Most studies so far have focused on IPL thickness as an indicator of its integrity, leading to contradictory findings, especially in glaucoma research.[44–47] Studies have shown that IPL thickness correlates with glaucoma diagnosis and advanced glaucoma,[45–47] while others have found no association between glaucoma diagnosis and IPL thinning,[54] or that RNFL and ganglion cell layer are better at distinguishing glaucomatous eyes.[55] This inconsistency stems from small sample sizes and variability within study populations. Factors such as age, ethnicity, sex, and common comorbidities like diabetes significantly influence retinal layer thickness and can act as confounders. However, many studies, including the present one, lack sufficient power to account for these variables due to their limited sample sizes.

Moreover, even if future studies confirm that IPL thinning does not precede ganglion cell layer thinning in neurodegenerative diseases like glaucoma, this does not imply that no early changes are occurring within this layer. Rather, it merely suggests that early dendritic changes in the IPL are not accompanied by thinning, at least initially. This underscores the need for a more comprehensive analysis of the IPL, which could include examining additional features, such as contrast variation and stratification of sublayers. These factors may serve as more effective biomarkers for early diagnosis and disease progression.

Our study has several strengths, including the use of commercially available OCT technology that enhances clinical accessibility and demonstrates strong repeatability in measurements both within and between eyes. We analyzed a substantial number of cases across a diverse range of age groups. However, we acknowledge important limitations. Our focus was on the IPL in a specific region, the nasal area from the fovea to the optic disc. Although we took several steps to minimize the effect of lens and vitreous changes on IPL contrast—such as z score normalization, the Contrast Limited Adaptive Histogram Equalization algorithm, and adjustment for image quality—we cannot fully exclude the possibility that optical media changes contributed to the IPL segmentation differences observed across age groups. Additionally, while we aimed to include a broad age range, the need for high-quality images free of artifacts made it more challenging to recruit older participants, who more frequently exhibited imaging limitations such as media opacities and unstable fixation. Most participants lacked axial length measurements, so results were not corrected for ocular magnification. Since participants having severe refractive errors (e.g., high myopia with a spherical equivalent equal or greater than −6.00 D) were excluded from the study, omitting axial length should not have significantly impacted our findings. To assess the potential effect of axial length on our measurements, we conducted a partial correlation analysis of age and IPL parameters, adjusting for axial length in a small subset of eyes with measured values. No significant changes in the strength or direction of correlations were found before and after this adjustment (see Appendix – Table A1, available at

www.ophthalmologyscience.org). In addition, since advanced OCT technologies used in prior studies are experimental and not commercially available, we were unable to perform a direct head-to-head comparison, although our findings align with previously reported IPL stratification patterns. Another limitation is that we did not fully automate the IPL segmentation, requiring manual selection of the region of interest.

In conclusion, the analysis of the IPL is both feasible and repeatable using commercial OCT technology. The IPL exhibits a characteristic intensity profile with 3 hyperreflective and 2 hyporeflective bands. With aging, not only does the IPL become thinner, but the contrast profile also flattens, resulting in less distinct hyperreflective and hyporeflective IPL sublayers. Increasing evidence from experimental studies suggests that synaptic alterations may occur before significant loss of axons and neurons, prompting a focus on the IPL as an early biomarker for diseases affecting synaptic health. However, previous studies have primarily concentrated on IPL thickness, which may not fully capture the early changes in the IPL associated with neurodegeneration. A more comprehensive assessment of the IPL, incorporating additional features such as those included in the present analysis and further understanding of the role of amacrine cells, could provide valuable insights for the early diagnosis and monitoring of neurodegenerative diseases.

Data Availability Statement

The data supporting the conclusions can be made available by the authors upon reasonable request.

Footnotes and Disclosures

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Analysis and interpretation: Correa, Emfietzoglou, Sakuno, Charonis, Vavvas

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Abbreviations and Acronyms:
D = diopters; **GCC** = ganglion cell complex; **HD** = high definition; **IPL** = inner plexiform layer; **MSE** = mean squared error; **RGC** = retinal ganglion cell; **RNFL** = retinal nerve fiber layer.

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References

1. Wässle H. Parallel processing in the mammalian retina. *Nat Rev Neurosci*. 2004;5:747–757.
2. Piccolino M, Strettoi E, Laurenzi E. Santiago Ramón Y Cajal, the retina and the neuron theory. *Doc Ophthalmol*. 1989;71: 123–141.
3. Marc RE. Neurochemical stratification in the inner plexiform layer of the vertebrate retina. *Vis Res*. 1986;26: 223–238.
4. Koontz MA, Hendrickson AE. Stratified distribution of synapses in the inner plexiform layer of primate retina. *J Comp Neurol*. 1987;263:581–592.
5. Dacey DM. Morphology of a small-field bistratified ganglion cell type in the macaque and human retina. *Vis Neurosci*. 1993;10:1081–1098.
6. Haverkamp S, Wässle H. Immunocytochemical analysis of the mouse retina. *J Comp Neurol*. 2000;424:1–23.
7. Wang J, O'Sullivan ML, Mukherjee D, et al. Anatomy and spatial organization of Müller glia in mouse retina. *J Comp Neurol*. 2017;525:1759–1777.
8. Haverkamp S, Haeseleer F, Hendrickson A. A comparison of immunocytochemical markers to identify bipolar cell types in human and monkey retina. *Vis Neurosci*. 2003;20:589–600.
9. Agostinone J, Di Polo A. Retinal ganglion cell dendrite pathology and synapse loss: implications for glaucoma. *Prog Brain Res*. 2015;220:199–216.
10. Almasieh M, Wilson AM, Morquette B, et al. The molecular basis of retinal ganglion cell death in glaucoma. *Prog Retin Eye Res*. 2012;31:152–181.
11. Alix JJP, Domingues AMde J. White matter synapses: form, function, and dysfunction. *Neurology*. 2011;76:397–404.
12. Hamos JE, DeGennaro LJ, Drachman DA. Synaptic loss in Alzheimer's disease and other dementias. *Neurology*. 1989;39: 355–361.
13. Puliafito CA, Hee MR, Lin CP, et al. Imaging of macular diseases with optical coherence tomography. *Ophthalmology*. 1995;102:217–229.
14. Dahrouj M, Saitakis G, Koulouri I, Vavvas DG. Optical coherence tomography. In: *Albert and Jakobiec's Principles and Practice of Ophthalmology*. Cham, Switzerland: Springer International Publishing; 2022:2713–2737.
15. Tanna H, Dubis AM, Ayub N, et al. Retinal imaging using commercial broadband optical coherence tomography. *Br J Ophthalmol*. 2010;94:372–376.
16. Zhang T, Kho AM, Srinivasan VJ. Improving visible light OCT of the human retina with rapid spectral shaping and axial tracking. *Biomed Opt Express*. 2019;10:2918–2931.
17. Miller DT, Kurokawa K. Cellular-scale imaging of transparent retinal structures and processes using adaptive optics optical coherence tomography. *Annu Rev Vis Sci*. 2020;6:115–148.
18. Zhang T, Kho AM, Srinivasan VJ. In vivo morphometry of inner plexiform layer (IPL) stratification in the human retina with visible light optical coherence tomography. *Front Cell Neurosci*. 2021;15:655096.
19. Lupidi M, Mangoni L, Centini C, et al. Quantitative and qualitative assessments of retinal structure with variable A-scan rate spectralis OCT: insights into IPL multilaminarity. *J Clin Med*. 2023;12:2637.
20. Schindelin J, Arganda-Carreras I, Frise E, et al. Fiji: an open-source platform for biological-image analysis. *Nat Methods*. 2012;9:676–682.
21. Sánchez-Sáez X, Ortuño-Lizarán I, Sánchez-Castillo C, et al. Starburst amacrine cells, involved in visual motion perception, lose their synaptic input from dopaminergic amacrine cells and degenerate in Parkinson's disease patients. *Transl Neurodegener*. 2023;12:17.
22. Chalupa LM, Günhan E. Development of on and off retinal pathways and retinogeniculate projections. *Prog Retin Eye Res*. 2004;23:31–51.
23. Norcia AM, Yakovleva A, Hung B, Goldberg JL. Dynamics of contrast decrement and increment responses in human visual cortex. *Transl Vis Sci Technol*. 2020;9:6.
24. Helmstaedter M, Briggman KL, Turaga SC, et al. Connectomic reconstruction of the inner plexiform layer in the mouse retina. *Nature*. 2013;500:168–174.
25. Sümbül U, Song S, McCulloch K, et al. A genetic and computational approach to structurally classify neuronal types. *Nat Commun*. 2014;5:3512.
26. Lauritzen JS, Anderson JR, Jones BW, et al. ON cone bipolar cell axonal synapses in the OFF inner plexiform layer of the rabbit retina. *J Comp Neurol*. 2013;521:977–1000.
27. Lin JB, Tsubota K, Apte RS. A glimpse at the aging eye. *NPJ Aging Mech Dis*. 2016;2:16003.
28. Zhou Y, Zhou W, Rao Y, et al. Dysregulated energy and protein homeostasis and the loss of GABAergic amacrine cells in aging retina. *Exp Eye Res*. 2024;245:109985.
29. Bonnel S, Mohand-Said S, Sahel JA. The aging of the retina. *Exp Gerontol*. 2003;38:825–831.
30. Skariah G, Todd PK. Translational control in aging and neurodegeneration. *WIREs RNA*. 2021;12:e1628.
31. Samuel MA, Zhang Y, Meister M, Sanes JR. Age-related alterations in neurons of the mouse retina. *J Neurosci*. 2011;31: 16033–16044.
32. Cuenca N, Ortuño-Lizarán I, Sánchez-Sáez X, et al. Interpretation of OCT and OCTA images from a histological approach: clinical and experimental implications. *Prog Retin Eye Res*. 2020;77:100828.
33. Kidd M. Electron microscopy of the inner plexiform layer of the retina in the cat and the pigeon. *J Anat*. 1962;96(Pt 2): 179–187.
34. Dowling JE, Boycott BB. Organization of the primate retina: electron microscopy. *Proc R Soc Lond B Biol Sci*. 1966;166:80–111.
35. Kolb H. Amacrine cells of the mammalian retina: neurocircuitry and functional roles. *Eye*. 1997;11:904–923.
36. Owsley C. Vision and aging. *Annu Rev Vis Sci*. 2016;2: 255–271.
37. Zandt BJ, Liu JH, Veruki ML, Hartveit E. AII amacrine cells: quantitative reconstruction and morphometric analysis of electrophysiologically identified cells in live rat retinal slices imaged with multi-photon excitation microscopy. *Brain Struct Funct*. 2017;222:151–182.
38. Roufail E, Rees S. Ageing has a differential effect on nitric oxide synthase-containing and catecholaminergic amacrine cells in the human and rat retina. *J Comp Neurol*. 1997;389: 329–347.
39. Rascunà C, Russo A, Terravecchia C, et al. Retinal thickness and microvascular pattern in early Parkinson's disease. *Front Neurol*. 2020;11:533375.
40. Liu M, Duggan J, Salt TE, Cordeiro MF. Dendritic changes in visual pathways in glaucoma and other neurodegenerative conditions. *Exp Eye Res*. 2011;92:244–250.
41. El-Danaf RN, Huberman AD. Characteristic patterns of dendritic remodeling in early-stage glaucoma: evidence from genetically identified retinal ganglion cell types. *J Neurosci*. 2015;35:2329–2343.
42. Henderson DCM, Vianna JR, Gobran J, et al. Longitudinal in vivo changes in retinal ganglion cell dendritic morphology

after acute and chronic optic nerve injury. *Invest Opthalmol Vis Sci*. 2021;62:5.
43. Ou Y, Jo RE, Ullian EM, et al. Selective vulnerability of specific retinal ganglion cell types and synapses after transient ocular hypertension. *J Neurosci*. 2016;36:9240−9252.
44. Ghassabi Z, Kuranov RV, Schuman JS, et al. In vivo sublayer analysis of human retinal inner plexiform layer obtained by visible-light optical coherence tomography. *Invest Opthalmology Vis Sci*. 2022;63:18.
45. Aydın R, Barış M, Durmaz-Engin C, et al. Early localized alterations of the retinal inner plexiform layer in association with visual field worsening in glaucoma patients. *PLoS One*. 2021;16:e024740.
46. Kim EK, Park HYL, Park CK. Segmented inner plexiform layer thickness as a potential biomarker to evaluate open-angle glaucoma: dendritic degeneration of retinal ganglion cell. *PLoS One*. 2017;12:e0182404.
47. Mohammadi M, Su E, Chew L, et al. Comparison of ganglion cell layer and inner plexiform layer rates of change in suspected and established glaucoma. *Am J Ophthalmol*. 2023;249:12−20.
48. Channa R, Lee K, Staggers KA, et al. Detecting retinal neurodegeneration in people with diabetes: findings from the UK Biobank. *PLoS One*. 2021;16:e0257836.
49. Ambiya V, Kumar A, Bhavaraj VR, et al. Study of inner retinal neurodegeneration in Diabetes Mellitus using spectral domain optical coherence tomography. *Eur J Ophthalmol*. 2022;32:3074−3081.
50. Shah J, Tan B, Wong D, et al. Evaluation of thickness of individual macular retinal layers in diabetic eyes from optical coherence tomography. *Sci Rep*. 2024;14:17909.
51. Ueda E, Hirabayashi N, Ohara T, et al. Association of inner retinal thickness with prevalent dementia and brain atrophy in a general older population: the hisayama study. *Ophthalmol Sci*. 2022;2:100157.
52. Kim JI, Kang BH. Decreased retinal thickness in patients with Alzheimer's disease is correlated with disease severity. *PLoS One*. 2019;14:e0224180.
53. Petzold A, Balcer LJ, Calabresi PA, et al. Retinal layer segmentation in multiple sclerosis: a systematic review and meta-analysis. *Lancet Neurol*. 2017;16:797−812.
54. Moghimi S, Fatehi N, Nguyen AH, et al. Relationship of the macular ganglion cell and inner plexiform layers in healthy and glaucoma eyes. *Transl Vis Sci Technol*. 2019;8:27.
55. Kim HJ, Lee SY, Park KH, et al. Glaucoma diagnostic ability of layer-by-layer segmented ganglion cell complex by spectral-domain optical coherence tomography. *Invest Opthalmol Vis Sci*. 2016;57:4799−4805.